Users' Privacy Perceptions and Data Disclosure Behaviors When Using AI Chatbots

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***Abstract*—AI chatbots such as ChatGPT and Claude have become integral productivity tools, yet users' understanding of how their conversational data is handled remains poorly understood. This mixed-methods study investigates users' privacy perceptions, mental models of AI data handling, and data disclosure behaviors through semi-structured interviews (n=10) and an online survey (n=47). Qualitative findings reveal four key themes: the privacy–convenience trade-off, widespread misconceptions about data handling, context-dependent disclosure decisions, and a gap between privacy knowledge and protective action. Survey results show that participants generally believe providers store conversations (M=3.51) and use data for training (M=3.64), yet remain uncertain about third-party access (M=2.60) and data sharing (M=2.34). Disclosure intentions were highest for professional tasks such as cover letters (M=3.94) and lowest for financial information (M=3.11). A significant positive correlation was found between privacy knowledge and disclosure willingness (ρ=0.303, p=0.038), suggesting that more knowledgeable users are paradoxically more willing to disclose. No significant differences were observed between technical and non-technical users. The most common protective behavior was avoidance-based (60%), while only 30% had opted out of data training. These findings highlight a critical need for transparent, user-friendly privacy controls and proactive disclosure warnings in AI chatbot interfaces.**

***Index Terms*—privacy, AI chatbots, data disclosure, mental models, user study**

I. Introduction

AI chatbots such as ChatGPT, Claude, and Gemini have rapidly become embedded in daily workflows, with millions of users sharing personal, professional, and sensitive information through these platforms. The conversational nature of these tools, combined with their human-like responsiveness, creates an environment where users may disclose more information than they would through traditional digital interfaces. This raises important questions about how users perceive the privacy implications of their interactions and whether their mental models of AI data handling align with actual practices.

Prior research has highlighted significant gaps in users' understanding of intelligent systems. Abdi et al. [1] found that users of smart home assistants frequently lack accurate mental models of data collection and processing. Zhang and Xu [2] demonstrated that perceived privacy risks significantly influence adoption willingness for AI-powered services, yet many users underestimate these risks. Ischen et al. [3] explored how the human-like qualities of conversational agents may encourage greater personal disclosure compared to traditional web forms. Malkin et al. [4] revealed significant misconceptions about data storage and access among smart speaker users, suggesting similar knowledge gaps likely exist for AI chatbot users. Despite growing interest in AI privacy, few studies have specifically examined how users' mental models of chatbot data handling compare to actual practices, or what factors drive disclosure decisions in conversational AI contexts.

This study aims to investigate four research questions: (RQ1) How do users perceive privacy risks when interacting with AI chatbots? (RQ2) What types of information do users disclose and in what contexts? (RQ3) Do users' mental models of AI data handling align with actual practices? (RQ4) What factors influence data disclosure decisions? We employ a mixed-methods approach combining semi-structured interviews (n=10) and an online survey (n=47) to provide both qualitative depth and quantitative breadth. Our findings contribute to the growing body of HCI research on AI privacy by documenting the persistence of the privacy paradox in conversational AI contexts, identifying previously underexplored dimensions of unintentional and third-party disclosure, and proposing design interventions grounded in empirical user data.

II. Related Work

A. The Privacy Paradox

The privacy paradox—the discrepancy between stated privacy concerns and actual disclosure behaviors—has been extensively documented in social media and e-commerce contexts [5]. Norberg et al. [5] found that individuals consistently disclose more information than their stated intentions predict. Kokolakis [9] provided a comprehensive review of the privacy paradox literature, identifying factors such as optimism bias, cognitive heuristics, and contextual pressures as contributing explanations. Whether this paradox extends to AI chatbot interactions, where the conversational format may reduce perceived formality and increase anthropomorphic trust, remains underexplored. Our study directly tests this relationship in the AI chatbot context by measuring both privacy knowledge and disclosure intentions.

B. Mental Models of Technology and Privacy

Mental models of technology significantly influence privacy behaviors. Wash [6] demonstrated that users' folk models of security threats directly shape their protective actions, even when these models are inaccurate. In the AI domain, Abdi et al. [1] found that smart speaker users held overly simplistic models of data flows, often believing data processing occurred locally rather than in the cloud. More recently, Liao and Sundar [10] found that users' understanding of how AI systems learn from data significantly influenced their comfort level with AI-powered services. These findings motivate our examination of how chatbot users conceptualize data storage, access, and retention, and whether the accuracy of these mental models predicts disclosure behavior.

C. Contextual Integrity and Disclosure Norms

The disclosure literature suggests that contextual factors strongly influence information sharing. Nissenbaum's theory of contextual integrity [7] posits that privacy violations occur when information flows deviate from context-specific norms. For AI chatbots, the relevant context is inherently ambiguous—users may frame their interaction as a private consultation, a professional tool, or a casual conversation, each carrying different disclosure norms. Ischen et al. [3] found that chatbots' anthropomorphic qualities blur these contextual boundaries, potentially encouraging disclosure that users would not make through other digital channels. Our study extends this line of inquiry by examining how different usage scenarios (professional, health, financial, personal) elicit different disclosure patterns and by documenting the prevalence of unintentional disclosure through indirect information sharing.

III. Methodology

This study employs a mixed-methods design consisting of two complementary studies. Each participant took part in only one study to avoid cross-contamination of responses and reduce participant burden. The study was cleared by the Carleton University Research Ethics Board (CUREB-B, Clearance #118932).

A. Study 1: Semi-Structured Interviews

We conducted semi-structured interviews with 10 participants who were regular AI chatbot users (at least once per week over the past six months). Participants were aged 18 or older and recruited through convenience sampling via personal networks, social media, and university mailing lists. Recruitment materials were framed generically as exploring "users' experiences and perceptions of AI chatbots" to avoid priming privacy concerns.

Interviews lasted 20–30 minutes and were conducted remotely via Zoom. With consent, sessions were audio-recorded and transcribed using Zoom's transcription feature; otherwise, written notes were taken. All 10 participants consented to audio recording. The interview protocol comprised 20 questions across five sections: (0) demographics, (1) general usage and experiences, (2) mental models of data handling, (3) data disclosure behaviors, and (4) privacy concerns and protective behaviors. Demographic information—including age range, gender, education level, and field of study or work—was collected at the beginning of each interview to contextualize participants' responses. The remaining questions were ordered from general to specific to allow privacy themes to emerge organically before directly prompting for privacy-related responses.

Table I summarizes participant demographics. The sample included 5 males, 4 females, and 1 non-binary participant, spanning ages 18–54, with diverse educational and professional backgrounds. Five participants had technical backgrounds (computer science, engineering, or IT-related fields).

TABLE I
Interview Participant Demographics

ID	Age	Gender	Education	Field	Tech
P1	25–34	M	Master's	Comp. Sci.	Yes
P2	18–24	F	Bachelor's	Psychology	No
P3	35–44	M	Doctoral	Elec. Eng.	Yes
P4	25–34	F	Master's	Marketing	No
P5	18–24	NB	Bachelor's	Design	No
P6	45–54	M	Master's	IT Mgmt.	Yes
P7	25–34	F	Bachelor's	Nursing	No
P8	18–24	M	Bachelor's	Comp. Eng.	Yes
P9	35–44	F	Master's	Social Work	No
P10	25–34	M	Master's	Data Sci.	Yes

Transcripts were analyzed using reflexive thematic analysis following Braun and Clarke's six-phase approach [8]. Both researchers independently coded approximately 20% of the transcripts (P1 and P3), achieving an inter-coder reliability of $\kappa = 0.78$ (Cohen's Kappa), indicating substantial agreement. Eight coding discrepancies were discussed and resolved collaboratively. After achieving satisfactory agreement, the remaining eight transcripts were divided between the two coders. Participants' mental models were systematically compared against documented practices from OpenAI's and Anthropic's published privacy policies to identify gaps in understanding.

B. Study 2: Online Survey

An online survey was administered via Qualtrics to 47 participants meeting the same eligibility criteria as Study 1. Survey participants did not participate in the interview study. The informed consent form was presented as the first page of the survey; participants were required to select "I agree to participate" before proceeding. The survey comprised six sections: (A) demographics, (B) usage patterns, (C) general perceptions (4 items, 5-point Likert scale from Strongly Disagree to Strongly Agree), (D) privacy knowledge beliefs (5 items, 5-point Likert scale), (E) scenario-based disclosure intentions (6 items, 5-point Likert scale from Very Unlikely to Very Likely), and (F) privacy protective behaviors (multi-select and open-ended). Section C was positioned before Section D to capture general attitudes before introducing privacy-specific items, thereby avoiding priming effects.

Survey data were analyzed using descriptive and inferential statistics in Python (pandas, scipy, matplotlib). Spearman's rank correlations examined relationships between privacy knowledge, general concerns, and disclosure intentions. Mann-Whitney U tests compared responses between participants with technical backgrounds (computer science, engineering, or IT; n=20) and non-technical backgrounds (n=27), using the self-reported field of study or work from Section A as the classification criterion.

IV. Results

A. Survey Demographics and Usage Patterns

The survey sample (n=47) was predominantly young adults: 45% aged 25–34, 30% aged 18–24, 11% aged 35–44, 4% aged 45–54, and 11% aged 55 or older. The sample comprised 55% males and 45% females. Most held graduate degrees (49% Master's, 40% Bachelor's, 9% high school). The largest professional group was Computer Science/Engineering/IT (43%), followed by Social Sciences/Humanities (15%), Business/Finance (11%), Arts/Design (9%), Health Sciences (6%), and Natural Sciences (2%), with 15% selecting "Other." Fig. 1 presents the demographic distributions.

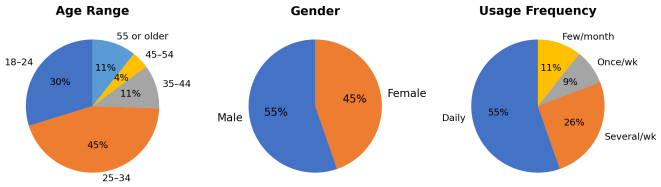

Fig. 1. Survey sample demographics: age range, gender, and usage frequency distributions (n=47).

Most participants were frequent users, with 55% using AI chatbots daily and 26% several times per week. ChatGPT was the most widely used platform (68%), followed by Gemini (38%), Claude (32%), and Grok (19%). The most common use cases were work/professional tasks (68%), information search (64%), academic work (51%), personal advice (40%), creative writing (34%), entertainment (30%), and coding/programming (28%).

B. Qualitative Findings: Interview Themes

Thematic analysis of the interview data produced 47 initial codes, which were iteratively grouped into 12 candidate themes and refined to four primary themes through collaborative discussion.

1) Theme 1: The Privacy–Convenience Trade-Off: All ten participants acknowledged privacy concerns related to AI chatbots, yet consistently prioritized productivity and convenience over these concerns in practice. This pattern was especially pronounced under time pressure. P6, an IT manager at a financial services firm, described sharing detailed network topology—firewalls, load balancers, subnet configurations, and error logs—with Claude during a production incident, acknowledging that doing so effectively provided "a detailed

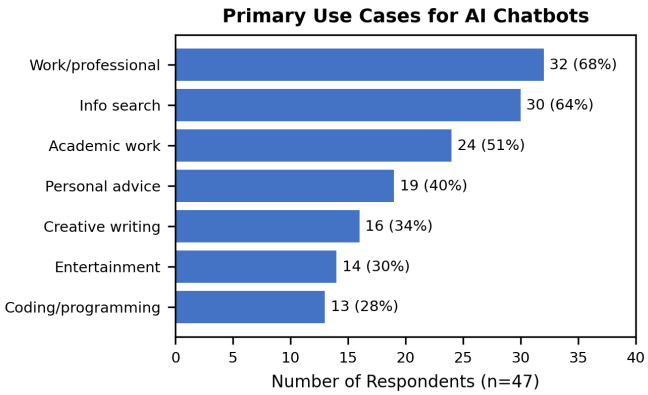

Fig. 2. Primary use cases for AI chatbots (n=47; multiple selections allowed).

blueprint of a financial institution's network infrastructure to a third-party AI service." P3 similarly noted that privacy precautions "degrade under time pressure, which is exactly when I'm sharing the most sensitive details," describing a "predictable gap between policy and practice." P10 shared 500 rows of real customer data for a churn prediction model, rationalizing it as "a small sample" before recognizing that "that reasoning doesn't hold up."

The trade-off was not limited to technical users. P9, a social worker, described extensive case debriefing sessions where she shared detailed information about vulnerable families, noting that ChatGPT "felt like the only space where I could be completely honest without consequences." P4 shared entire product briefs without considering confidentiality until a colleague intervened. These findings suggest that the convenience calculus operates independently of technical sophistication, driven instead by immediate task demands and emotional states.

2) Theme 2: Misconceptions and Knowledge Gaps: A stark divide emerged between technical and non-technical participants' mental models, though neither group's understanding consistently predicted appropriate behavior. Technical participants (P1, P3, P6, P8, P10) demonstrated sophisticated understanding of cloud infrastructure, distributed systems, and data persistence. P3 accurately described conversation storage across cloud infrastructure with separate training pipelines and recognized that "encryption is necessary, not sufficient—the real question is key management policy and access control implementation." P6 articulated the distinction between user-facing deletion and actual data purging across backups and analytics pipelines.

Non-technical participants held significantly vaguer mental models. P2 described data storage as "somewhere on the internet" and assumed conversations were "like a private diary," expressing surprise at the possibility of company access. P9 had not considered that employees could access conversations and became visibly distressed during the interview: "I assumed it was just me. [...] But the company can see it too, can't they? [...] I've shared client details. Not names, but... oh."

P5 initially expressed no privacy concerns, stating "I feel like I'm not interesting enough for anyone to care about my data," reflecting a fundamental misunderstanding of how aggregated conversational data creates value regardless of individual user "interestingness."

A critical finding was that even accurate knowledge did not guarantee appropriate behavior. P3, despite understanding training data practices in detail, had not disabled the training toggle on ChatGPT for six months. P10 had not checked Claude's data handling settings despite sharing customer data through the platform. P1 acknowledged being unaware of Claude's consumer data policies despite regular use. This disconnect between knowledge and action was observed across the technical subsample.

3) Theme 3: Context-Dependent and Unintentional Disclosure: Participants' disclosure behaviors varied dramatically by context. Professional and task-oriented sharing was perceived as lower-risk than personal disclosure. P4 shared entire product briefs and competitive intelligence without hesitation, while P7 described patient medication regimens without considering confidentiality implications until an instructor raised the issue. Conversely, P2 made mental health disclosures hypothetical ("if someone was dealing with X") out of embarrassment rather than privacy concern.

A particularly significant finding was the prevalence of unintentional information disclosure through indirect means. Participants frequently revealed sensitive information without recognizing it as disclosure: location through restaurant queries (P1: "I'll say 'restaurants near Byward Market' without thinking"), institutional identity through academic questions (P2: "I'll literally be like 'I'm a third-year psych student at uOttawa' "), infrastructure details through technical queries (P6: "every technical question reveals infrastructure detail"), and research trajectories through query patterns (P3: "my query patterns map my research trajectory"). P4 articulated the aggregation risk: individual queries "feel like nothing individually but add them up and it's basically a business profile."

Third-party disclosure without consent emerged as a previously underexplored dimension. P8 shared a friend's private messages to evaluate an argument, P3 shared a collaborator's email content without consent, P9 described vulnerable clients' case details, and P7 provided patient information that could enable re-identification in a small clinical setting. These disclosures extend the privacy implications of AI chatbots beyond individual users to their social and professional networks.

4) Theme 4: Gap Between Awareness and Action: A consistent gap between privacy awareness and protective behavior emerged across the sample. Awareness of privacy settings was low among non-technical users: P2, P5, P7, and P9 were entirely unaware of training opt-out features. P5's response was representative: "Wait—there's a way to opt out of the training thing? How?" P9 similarly stated: "I had no idea. I just type and send."

Even among aware users, adoption of protective measures was inconsistent. P1 described "keep[ing] meaning to" disable training for six months. P6, despite thorough knowledge of available options and professional expertise in IT security, acknowledged a persistent gap between intention and practice under operational pressure. P8, who was fully aware of all settings, reported having used none of them, expressing indifference: "Don't care enough to toggle anything off."

Protective behaviors were predominantly triggered by external events rather than proactive concern. P4 began using placeholder names only after a CTO intervention; P7 modified patient details only after an instructor's warning; P10 adopted synthetic data practices only after the discomfort of sharing real customer records. P9's interview itself served as a triggering event: "This conversation is showing me I need to completely change my approach." This pattern suggests that the awareness–action gap may be closed not by education alone but by salient, contextually relevant prompts.

C. Quantitative Findings: Survey Results

1) General Perceptions: Participants reported moderate comfort sharing personal details (M=3.13, SD=0.82) and moderate trust in data handling (M=3.30, SD=0.83). Privacy concerns were slightly elevated (M=3.40, SD=0.77), and perceived control over shared information was moderate (M=3.38, SD=0.77). The overall pattern suggests ambivalence: participants neither strongly trust nor strongly distrust AI chatbot data practices. Notably, for all four items the modal response was "Neutral," indicating that a substantial proportion of users have not formed strong opinions about these issues.

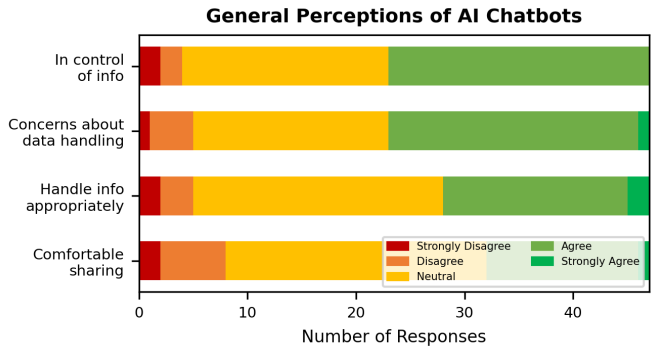

Fig. 3. Distribution of general perception responses (n=47). Items rated on 5-point Likert scale from Strongly Disagree to Strongly Agree.

2) Privacy Knowledge: Participants showed relatively strong agreement that conversations are used for AI training (M=3.64, SD=0.85) and that providers store conversation history by default (M=3.51, SD=0.80). Belief in the ability to delete conversation history was moderate but with high variance (M=3.62, SD=1.03), suggesting divergent beliefs about deletion effectiveness. Participants were notably uncertain about third-party data access (M=2.60, SD=1.10) and whether providers share data with other companies (M=2.34, SD=1.05). This pattern indicates reasonable awareness of direct, first-party data practices but significant uncertainty about downstream data flows involving third parties—a finding that

aligns with interview participants' difficulty articulating what happens to their data beyond the immediate provider.

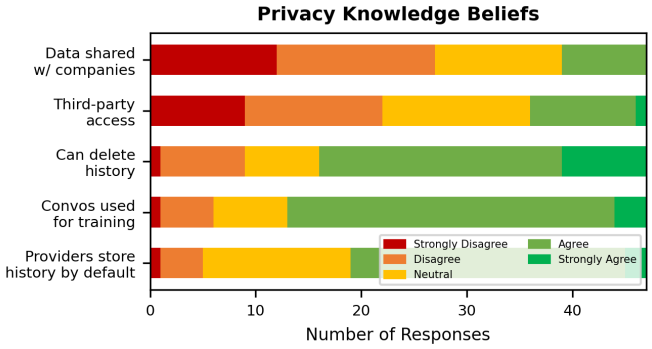

Fig. 4. Distribution of privacy knowledge belief responses (n=47). Items rated on 5-point Likert scale.

3) Disclosure Intentions: The highest willingness to disclose was for drafting cover letters with work history (M=3.94, SD=0.87), followed by health symptoms (M=3.85, SD=1.10) and proprietary work documents (M=3.70, SD=0.93). Lower but still moderate willingness was reported for location-based recommendations (M=3.47, SD=1.06), relationship conflicts (M=3.43, SD=0.97), and financial information (M=3.11, SD=1.07). The finding that proprietary work documents received relatively high disclosure willingness (M=3.70) despite their objective sensitivity is notable and consistent with interview themes around professional contexts normalizing disclosure. Financial disclosure showed the lowest mean and highest resistance, which aligns with the perception of financial data as uniquely sensitive.

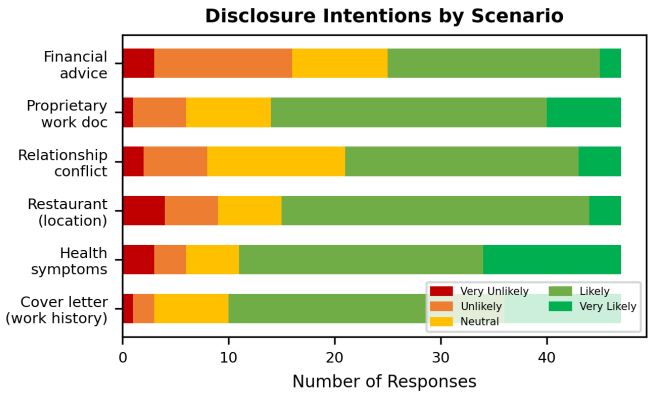

Fig. 5. Distribution of disclosure intention responses by scenario (n=47). Items rated on 5-point scale from Very Unlikely to Very Likely.

4) Privacy Protective Behaviors: The most common behavior was avoidance-based: 60% of respondents reported having avoided sharing certain information. Active protective measures were less common: 43% had deleted chat history, 38% had adjusted privacy settings, 34% had used incognito or anonymous mode, and only 30% had opted out of data training. Just 11% reported taking no protective actions at all. This pattern corroborates the interview finding that passive avoidance is the default protective strategy, while active engagement with privacy settings remains a minority behavior.

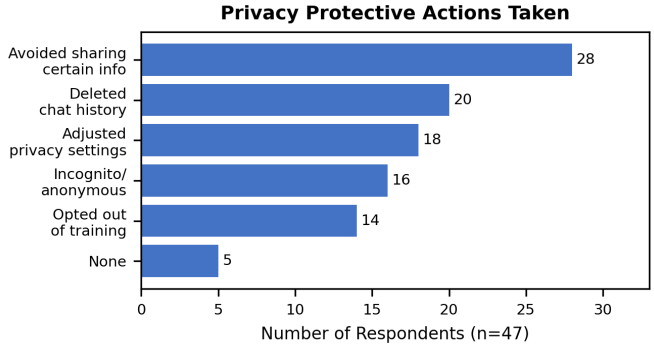

Fig. 6. Frequency of privacy protective actions taken by survey respondents (n=47; multiple selections allowed).

5) Correlation Analysis: Spearman's rank correlation revealed a statistically significant positive correlation between average privacy knowledge scores and average disclosure intention scores ($\rho = 0.303$, $p = 0.038$). This counterintuitive finding suggests that participants with greater privacy knowledge were *more*, not less, willing to disclose information in the presented scenarios. No significant correlation was found between privacy concerns (General Perceptions item 3: "I have concerns about how AI chatbot providers handle the data from my conversations") and disclosure intentions ($\rho = 0.102$, $p = 0.494$), indicating that expressed concern does not translate into reduced disclosure willingness.

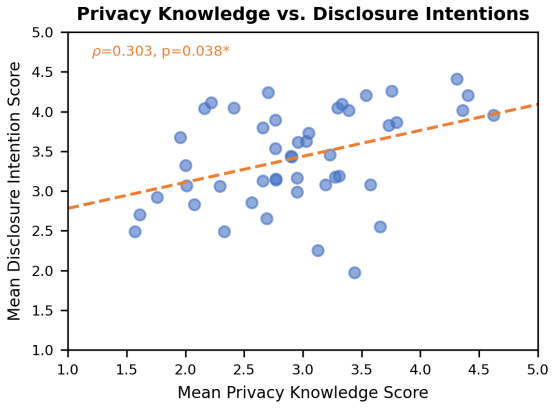

Fig. 7. Scatter plot of mean privacy knowledge scores vs. mean disclosure intention scores (n=47) with trend line. Spearman's $\rho = 0.303$, $p = 0.038$.

6) Technical vs. Non-Technical Group Comparisons: Mann-Whitney U tests revealed no statistically significant differences in privacy knowledge ($U = 227.5$, $p = 0.363$; technical M=3.06, non-technical M=3.20), disclosure intentions ($U = 213.5$, $p = 0.224$; technical M=3.46, non-technical M=3.67), or general perceptions ($U = 291.0$, $p = 0.654$; technical M=3.36, non-technical M=3.26). While the interview

data revealed clear qualitative differences in *how* technical and non-technical users articulated their understanding (detailed infrastructure knowledge vs. vague "cloud" references), these differences did not translate into statistically distinct survey response patterns. This may reflect a ceiling effect in the Likert-scale items, which capture agreement strength but not the depth or accuracy of underlying reasoning.

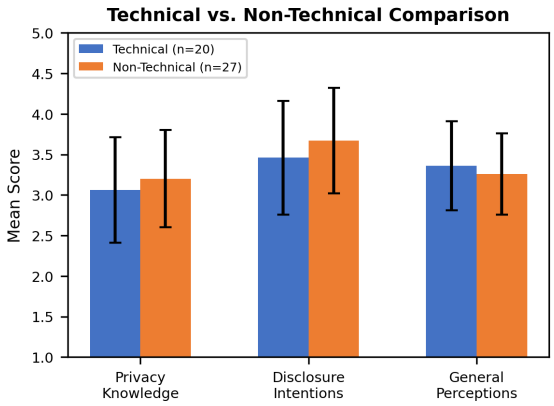

Fig. 8. Comparison of mean scores between technical (n=20) and non-technical (n=27) participants. Error bars represent standard deviations. No significant differences (Mann-Whitney U, all $p > 0.05$).

V. Discussion

A. The AI Privacy Paradox

Our findings extend the well-documented privacy paradox [5] into the AI chatbot context with a notable twist. Both interview and survey data demonstrate that privacy awareness does not reduce disclosure—and may even increase it. The significant positive correlation between privacy knowledge and disclosure intentions ($\rho = 0.303$, $p = 0.038$) suggests that more knowledgeable users may feel better equipped to evaluate and accept risks, leading to greater willingness to share through informed trade-offs rather than ignorance. This interpretation is strongly supported by interview findings: technically sophisticated participants (P1, P3, P6, P10) explicitly described making calculated decisions to share sensitive data after weighing productivity benefits against perceived risks. P3 characterized this as being "acutely aware" of information leakage while continuing "because the productivity benefit is substantial." P6 described his sharing as "calculated productivity decisions that I probably wouldn't defend under scrutiny."

The absence of a significant correlation between privacy concerns and disclosure ($\rho = 0.102$, $p = 0.494$) further supports this interpretation: expressed concern exists in a separate cognitive domain from behavioral decisions. This finding aligns with Kokolakis's [9] analysis of the privacy paradox, which identifies cognitive heuristics and contextual pressures as mediating factors between attitudes and behaviors. In the AI chatbot context, the immediacy of task demands and the perceived "private conversation" framing may particularly weaken the attitude–behavior link.

B. Mental Model Gaps and Their Consequences

The qualitative data revealed that mental model accuracy varied substantially but consistently failed to predict protective behavior. Technical participants provided detailed and largely accurate descriptions of cloud infrastructure, data pipelines, and access control mechanisms yet continued sharing sensitive information. The critical disconnect was not between knowledge and concern but between concern and action. As P3 noted: "awareness doesn't translate perfectly to action. Convenience often wins, especially at midnight when I'm chasing a deadline."

Non-technical participants operated with vague but not necessarily more harmful mental models—their lower disclosure in some contexts reflected general caution rather than informed risk assessment. P2's "private diary" assumption, while inaccurate, led her to treat the chatbot as a safe confidant, resulting in extensive emotional disclosure. P9's complete unawareness of data practices led to professional ethics concerns when she realized client information had been shared with a commercial AI system.

A particularly concerning finding was the widespread uncertainty about data persistence. The survey showed that belief in deletion capability was moderate but highly variable (M=3.62, SD=1.03—the highest standard deviation among privacy knowledge items). Interview participants across technical backgrounds questioned whether "delete" meant actual data purging or merely UI removal. P1 noted that "in distributed systems, real deletion is a pain," while P3 estimated "months to years for derived data, even post-deletion." This uncertainty about the permanence of shared information has significant implications for informed consent frameworks, which typically assume users can meaningfully evaluate data handling practices.

C. Contextual Norms, Unintentional Disclosure, and Third-Party Risk

Consistent with Nissenbaum's contextual integrity framework [7], disclosure decisions were heavily context-dependent. Professional contexts (cover letter drafting, M=3.94; work documents, M=3.70) elicited higher disclosure willingness than personal contexts (financial advice, M=3.11; relationship conflicts, M=3.43). This hierarchy suggests that users apply a "professional tool" frame that normalizes information sharing in work contexts while maintaining greater guardedness in personal domains. The interview data further revealed that professional norms can override privacy considerations entirely: P4 did not consider product strategy information as privacy-sensitive until a colleague explicitly flagged the concern.

The prevalence of unintentional disclosure represents a significant challenge that current consent-based privacy frameworks do not adequately address. Unlike deliberate sharing, unintentional disclosure through location references, institutional affiliations, and aggregated query patterns cannot be mitigated through user education alone. P6's observation that "every technical question reveals infrastructure detail" and P3's awareness that queries "collectively reveal current

projects, methodological gaps, even funding sources" highlight how the conversational format of chatbots encourages a level of contextual detail that users do not recognize as information disclosure.

Third-party disclosure emerged as a particularly under-explored dimension. When P8 shared a friend's messages, P3 shared a collaborator's email, P9 described clients' case details, and P7 provided identifiable patient information, the privacy implications extended to individuals who had no knowledge of or consent to their information entering an AI system. This finding has significant implications for professional contexts involving confidential information, particularly in healthcare (PHIPA), social work, and legal practice.

D. Design Implications

Based on our findings, we propose five design recommendations. First, real-time sensitivity detection: automated systems that detect and flag potentially sensitive information—personal health data, financial details, third-party identifying information, proprietary business content—before submission, addressing both intentional and unintentional disclosure. P5's suggestion of a "button before you send something personal that's like 'this is sensitive, please don't keep it' " captures this need. Second, graduated privacy modes with visual indicators: clear, persistent visual signals (as P4 suggested, "green for private, red for 'this may be used' ") indicating the current data handling status of a conversation, reducing the cognitive burden of tracking privacy implications. Third, transparent data dashboards: user-facing interfaces showing stored data and its uses, as requested by P1 and P10, enabling users to audit their information footprint. Fourth, prominent, simplified privacy controls: settings surfaced in the main interface rather than buried in menus, as P2 and P5 noted they would never proactively seek out privacy settings. Fifth, profession-specific compliance modes: tailored data handling for users in regulated professions (healthcare, social work, legal, financial services), with verifiable guarantees appropriate to their regulatory obligations.

E. Limitations and Future Work

This study has several limitations. The convenience sampling method limits generalizability, and the sample skewed toward young, highly educated users with technical backgrounds. The survey sample size (n=47) limits statistical power, particularly for subgroup analyses. Self-reported disclosure intentions may not perfectly reflect actual behaviors, a limitation inherent to survey-based privacy research. The interview sample (n=10), while providing rich qualitative data, may not capture the full range of user experiences. Additionally, the study was conducted with English- and Chinese-speaking users primarily in a Canadian university context, which may not represent global user populations or less tech-savvy demographics.

Future research should employ larger, more diverse samples and consider behavioral observation methods—such as analyzing actual chatbot conversation logs with consent—to complement self-report data. Longitudinal studies could examine how privacy perceptions and behaviors evolve over time, particularly in response to data breaches, policy changes, or privacy-relevant media coverage. The positive correlation between privacy knowledge and disclosure warrants further investigation to disentangle whether this reflects informed risk acceptance, optimism bias, or a third mediating factor such as technology self-efficacy.

VI. Conclusion

This study provides mixed-methods evidence that AI chatbot users face a complex privacy landscape characterized by incomplete mental models, context-dependent disclosure decisions, and a persistent gap between privacy awareness and protective action. Our findings reveal that the privacy paradox extends to AI chatbot interactions: greater privacy knowledge correlates with increased, not decreased, disclosure willingness ($\rho = 0.303$, $p = 0.038$). Users across technical backgrounds share sensitive personal, professional, and third-party information, often unintentionally, while protective behaviors remain predominantly passive (60% avoidance-based) rather than active (30% training opt-out). Technical and non-technical users differ qualitatively in their understanding but not quantitatively in their behaviors. These findings underscore the need for AI chatbot providers to move beyond consent-based privacy frameworks toward proactive, system-level privacy protections that support users in making informed disclosure decisions in real time, with particular attention to unintentional disclosure and third-party information that current designs fail to address.

Acknowledgment

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APPENDIX

- **Fangyuan Lin:** Study design, interview protocol development, conducted 5 interviews (P1, P3, P5, P7, P9), qualitative coding, survey design and deployment on Qualtrics, quantitative data analysis (descriptive and inferential statistics), data visualization, report writing (Introduction, Methodology, Results, Discussion).
- **Jingyan Jiang:** Literature review, interview protocol development, conducted 5 interviews (P2, P4, P6, P8, P10), qualitative coding and inter-coder reliability assessment, survey recruitment coordination, report writing (Related Work, Conclusion, Appendices), presentation preparation.

A. Recruitment Invitation – Interview Study

Volunteers needed for a research study on AI chatbot use.

We are looking for volunteers for a research study exploring users' experiences and perceptions when using AI chatbots (e.g., ChatGPT, Claude, Gemini). This project is part of a graduate course SYSC5009: Human-centric Security and Privacy at Carleton University.

If you participate, you will take part in a brief interview (20–30 minutes) about your experiences with AI chatbots. With your consent, the interview will be audio-recorded to help with data analysis. Audio-recording is optional.

To be eligible, you must be 18 years or older and have used an AI chatbot at least once per week over the past six months.

The interview can be completed via video-conference. There is no compensation for participating.

If you are interested, please email Fangyuan Lin at fangyuanlin@cmail.carleton.ca for more details.

This research has been cleared by Carleton University Research Ethics Board CUREB-B Clearance #: 118932. Should you have any ethical concerns, please contact the REB Chair at ethics@carleton.ca.

B. Recruitment Invitation – Survey Study

Volunteers needed for a research study on AI chatbot use.

We are looking for volunteers to complete a short online survey (10–15 minutes) about your experiences and perceptions when using AI chatbots (e.g., ChatGPT, Claude, Gemini). This project is part of a graduate course SYSC5009: Human-centric Security and Privacy at Carleton University.

To be eligible, you must be 18 years or older and have used an AI chatbot at least once per week over the past six months.

There is no compensation for participating. The survey will be completed online via Qualtrics.

If you are interested, please email Fangyuan Lin at fangyuanlin@cmail.carleton.ca to receive the survey link.

This research has been cleared by Carleton University Research Ethics Board CUREB-B Clearance #: 118932. Should you have any ethical concerns, please contact the REB Chair at ethics@carleton.ca.

C. Informed Consent Form – Interview Study

Name and Contact Information of Researchers:
Fangyuan Lin, Carleton University
Jingyan Jiang, Carleton University
Email: fangyuanlin@cmail.carleton.ca / jingyanjiang@cmail.carleton.ca

Supervisor: Prof. Hala Assal, Department of Systems and Computer Engineering, Carleton University (Hala.Assal@carleton.ca)

Title: SYSC5009 – Human-centric Security and Privacy Project: Users' Experiences and Perceptions When Using AI Chatbots

Carleton University Project Clearance: Clearance #: 118932, Date of Clearance: Jan 9, 2026

This project is being completed as part of SYSC5009, a graduate course at Carleton University. This study aims to explore users' experiences and perceptions when using AI chatbots such as ChatGPT and Claude. We aim to understand how users interact with these tools and what factors shape their experiences.

Participation in this study is completely voluntary. There is no obligation to participate.

The study involves one interview session lasting approximately 20–30 minutes. You will be asked about your experiences using AI chatbots, how you use these tools, and your general perceptions of them. Demographic information will also be collected.

If you provide explicit consent at the end of this form, the researcher will audio-record this session to help with data analysis. We will use Zoom transcription to transcribe the recording. The audio recording and transcription will only be used for this purpose. If you do not wish to be audio-recorded, the researcher will take written notes instead.

Your data will be kept confidential and associated with an anonymous code (e.g., P1, P2). All research data will be password-protected and accessible only by the researchers and supervisor. Data will be securely destroyed after grades have been finalized.

Risks and Inconveniences: We do not anticipate any risks to participating in this study.

Compensation: There is no compensation for participation.

Ethics Review: This project was reviewed and cleared by the Carleton University Research Ethics Board B. If you have any ethical concerns, please contact ethics@carleton.ca.

Statement of Consent:
I voluntarily agree to participate in this study. ☐ Yes ☐ No
I agree to be audio recorded and transcription through Zoom. ☐ Yes ☐ No
Signature of participant: _____ Date: _____
Signature of researcher: _____ Date: _____

D. Informed Consent Form – Survey Study

Name and Contact Information of Researchers:
Fangyuan Lin, Carleton University

Jingyan Jiang, Carleton University
Email: fangyuanlin@cmail.carleton.ca / jingyanjiang@cmail.carleton.ca

Supervisor: Prof. Hala Assal, Department of Systems and Computer Engineering, Carleton University (Hala.Assal@carleton.ca)

Title: SYSC5009 – Human-centric Security and Privacy Project: Users' Experiences and Perceptions When Using AI Chatbots

Carleton University Project Clearance: Clearance #: 118932, Date of Clearance: Jan 9, 2026

This project is being completed as part of SYSC5009, a graduate course at Carleton University. This study aims to explore users' experiences and perceptions when using AI chatbots such as ChatGPT and Claude. We aim to understand how users interact with these tools and what factors shape their experiences.

Participation in this study is completely voluntary. There is no obligation to participate.

The study involves completing an online survey that takes approximately 10–15 minutes. You will be asked about your experiences using AI chatbots, how you use these tools, and your general perceptions of them. Demographic information will also be collected.

Your responses will be anonymous. All research data will be password-protected and accessible only by the researchers and supervisor. Data will be securely destroyed after grades have been finalized.

Risks and Inconveniences: We do not anticipate any risks to participating in this study.

Compensation: There is no compensation for participation.

Ethics Review: This project was reviewed and cleared by the Carleton University Research Ethics Board B. If you have any ethical concerns, please contact ethics@carleton.ca.

By clicking "I agree to participate" below, you confirm that you have read and understood the above information and voluntarily consent to participate in this survey.

☐ I agree to participate ☐ I do not agree to participate

Survey: Users' Experiences and Perceptions of AI Chatbots

[Hosted on Qualtrics. Consent form displayed on first page; participants must click "I agree to participate" before proceeding.]

Section A: Demographics

1) What is your age range? [18–24 / 25–34 / 35–44 / 45–54 / 55+]
2) What is your gender? [Male / Female / Non-binary / Prefer not to say]
3) What is your highest level of education? [High school / Bachelor's / Master's / Doctoral / Other]
4) What is your field of study or work? [Computer Science/Engineering/IT / Natural Sciences / Social Sciences/Humanities / Business/Finance / Health Sciences / Arts/Design / Other: ___]
5) How frequently do you use AI chatbots? [Daily / Several times a week / Once a week / A few times a month]

Section B: Usage Patterns

1) Which AI chatbots have you used? (Select all that apply) [ChatGPT / Claude / Gemini / Grok / Other: ___]
2) What do you primarily use AI chatbots for? (Select all that apply) [Work/professional tasks / Academic work / Creative writing / Coding/programming / Personal advice / Information search / Entertainment / Other: ___]

Section C: General Perceptions and Concerns (5-point Likert: Strongly Disagree to Strongly Agree)

Prompt: Please indicate the extent to which you agree with the following statements about your experience with AI chatbots.

1) I feel comfortable sharing personal details during conversations with AI chatbots
2) I think AI chatbots understand and handle my information appropriately
3) I have concerns about how AI chatbot providers handle the data from my conversations
4) I feel in control of the information I share when using AI chatbots

Section D: Privacy Knowledge (5-point Likert: Strongly Disagree to Strongly Agree)

Prompt: Please indicate the extent to which you agree with the following statements about how AI chatbots handle your data.

1) AI chatbot providers store my conversation history by default
2) My conversations may be used to improve or train AI models
3) I can delete my conversation history from the AI chatbot provider's servers
4) Third parties (e.g., advertisers, researchers) may access anonymized conversation data
5) AI chatbot providers share my data with other companies

Section E: Data Disclosure Intentions (5-point Likert: Very Unlikely to Very Likely)

Prompt: For each scenario below, please rate how likely you would be to proceed with sharing the described information with an AI chatbot.

1) You are asking an AI chatbot to help you draft a cover letter and it asks for your work history and skills.
2) You are describing physical symptoms to an AI chatbot to get health advice.
3) You are asking an AI chatbot to recommend restaurants by providing your current neighborhood or address.
4) You are sharing details about a personal relationship conflict to get advice from an AI chatbot.
5) You are pasting a work document containing proprietary information into an AI chatbot for editing.
6) You are asking an AI chatbot for financial advice by sharing your income and expenses.

Section F: Privacy Protective Behaviors

1) Have you ever taken any of the following actions? (Select all that apply) [Deleted chat history / Used incognito or anonymous mode / Adjusted privacy settings / Opted out of data training / Avoided sharing certain information / None of the above]
2) Is there anything else you would like to share about your experience with AI chatbots? [Open-ended text box]

The semi-structured interview comprised 20 questions across five sections:

Section 0 – Demographics:

1) How old are you?
2) What is your gender?
3) What is your highest level of education?
4) What is your field of study or work?

Section 1 – General Usage and Experiences:

5) Which AI chatbots do you use, and how frequently?
6) What tasks do you typically use AI chatbots for?
7) Can you walk me through a recent experience where you had a particularly useful or memorable interaction with an AI chatbot? What information did you share during that interaction?
8) What concerns, if any, do you have about using AI chatbots in general?

Section 2 – Mental Models of Data Handling:

9) Where do you think your conversations with AI chatbots are stored?
10) Who do you think can access your chat history?
11) Do you believe your conversations are used to train AI models? Why or why not?
12) How long do you think your data is retained by the chatbot provider?

Section 3 – Data Disclosure Behaviors:

13) Can you describe a specific situation where you ended up sharing personal or sensitive information (e.g., health concerns, financial details, relationship issues) with an AI chatbot? What prompted you to share that information?
14) Have you ever shared information with an AI chatbot that you later regretted or felt uncomfortable about? What happened?
15) Sometimes users share personal details indirectly—for example, asking "suggest restaurants near [home address]" or describing symptoms. Have you had experiences like this? How aware were you of the information you were revealing?
16) Have you ever hesitated to ask something or changed how you phrased a question due to concerns about sharing information? Can you give an example?

Section 4 – Privacy Concerns and Protective Behaviors:

17) What privacy concerns, if any, do you have about using AI chatbots specifically?
18) Are you aware of any privacy settings, data deletion options, or opt-out features offered by AI chatbot providers? Have you used any of them?
19) What steps, if any, do you take to protect your privacy when using these tools?
20) What changes would make you feel more comfortable sharing information with AI chatbots?

Note: Demographic questions are collected first to contextualize responses. The remaining questions are ordered from general to specific to avoid priming participants on privacy topics and allow privacy concerns to emerge organically.

Full transcripts for all 10 interview participants are presented below. Each transcript is organized by interview section and question.

E. Participant P1 – Computer Science, Male, 25–34

Demographics:

Q1: How old are you? 28.

Q2: What is your gender? Male.

Q3: What is your highest level of education? Master's — did my M.Sc. in comp sci.

Q4: What is your field of study or work? Computer science. I'm a software developer at a mid-size tech company.

General Usage:

Q5: Which AI chatbots do you use, and how frequently? ChatGPT every day, it's basically a tab that never closes. Claude maybe three, four times a week — I like it more for like, longer back-and-forth coding sessions. And Copilot's always on in VS Code, but I don't really think of that as a chatbot, it's more like autocomplete on steroids. So yeah, total interactions... probably ten to fifteen a day? More if something's broken.

Q6: What tasks do you typically use AI chatbots for? Coding stuff, like 80% of the time. Debugging, boilerplate, 'hey why is this test failing.' The rest is... I dunno, random things. Trip planning, recipes. I once spent like 20 minutes asking it about the best way to organize a closet. My girlfriend thought that was hilarious.

Q7: Memorable interaction? Oh yeah, so last week I had this auth issue in a microservices thing at work. JWT tokens weren't propagating through the API gateway correctly. I just dumped the whole config — service names, endpoint URLs, the error logs, everything — into Claude and was like 'fix this.' And it actually spotted the problem, some header that was getting stripped at the gateway. Saved me like two hours. [pause] I mean, in retrospect I shared a LOT of internal stuff. Service architecture, naming conventions... but in the moment you just want the thing to work, you know?

Q8: General concerns? Accuracy, mainly. Sometimes the code looks right but it's subtly wrong, and if you're not paying attention you'll ship a bug that came from the AI. Privacy... I think about it sometimes? Like when I paste work code I get a little twinge. But my company doesn't even have a policy on it, so I figure it's not that big a deal.

Mental Models:

Q9: Where are conversations stored? Their cloud. OpenAI's on Azure, Anthropic probably AWS or GCP. Standard stuff — database for the conversation objects, probably some blob

storage, user account tables. It's all there because I can see my history. I don't think anything's local, maybe some browser cache.

Q10: Who can access your chat history? Me, obviously. Then... some engineers at the company, trust and safety people. I'd hope there's RBAC and audit logs and all that, but honestly who knows. I'd be surprised if random employees can just browse chats, but technically the capability is there for someone with DB access.

Q11: Used for training? ChatGPT, yeah, unless you toggle that setting off. Which I haven't done because I keep forgetting. Claude — I think Anthropic says they don't? Or at least not the API. I don't actually know the details for the consumer product. I should probably look into that. [laughs] I've been saying 'I should look into that' for like six months.

Q12: Data retention? No clue. Probably a long time. Like, I can delete a conversation from my sidebar, but does that actually nuke it from their databases? From the backups? Probably not. In distributed systems, real deletion is a pain. I'd bet it lingers for months or years in some form.

Disclosure Behaviors:

Q13: Sensitive information shared? The work code thing is the big one — API endpoints, DB schemas, once I left test credentials in a snippet by accident. That was an 'oh shit' moment, I rotated them right away but still. And uh, I asked ChatGPT about salary negotiation once and gave it my actual numbers. Current comp, the offer, equity breakdown. That felt weird after.

Q14: Regretted sharing? The credentials, 100%. Even though they were staging creds. And the salary thing — I could've just asked 'how do you negotiate a 20% raise' without giving my actual numbers. Didn't need to be that specific.

Q15: Indirect sharing? Yeah, like I'll say 'restaurants near Byward Market' without thinking — that's my location right there. Or the code I paste, the project names and conventions reveal what my company builds. You don't think of it as 'sharing personal info' but it totally is.

Q16: Hesitated or changed phrasing? A bit. When I was thinking about switching jobs, I wanted to ask about a specific company's interview but I got paranoid — irrational, I know — and asked generically instead. And I've started stripping company names from code before pasting. Not consistently, but when I remember.

Privacy Concerns:

Q17: Specific privacy concerns? The work code thing. If someone at OpenAI can see what I've pasted, they could piece together our product architecture. Data breaches too — if these companies get popped, that's a treasure trove of people's code, conversations, everything. And the training thing bugs me. I don't want my code showing up in someone else's Copilot suggestions.

Q18: Aware of privacy settings? Yeah, ChatGPT has the toggle to disable training. I keep meaning to turn it on. There's also the 'temporary chat' thing I think? I never use it. Claude, I honestly haven't explored the settings at all. I know you can delete convos but that's about it.

Q19: Protective steps? I try to strip sensitive stuff from code — company names, real URLs. 'Try' being the operative word, I'm not great about it. I use a personal account instead of work email, which probably doesn't help at all. And I've been using Claude more for work because I have this vague sense Anthropic is more privacy-friendly. No real evidence for that, just vibes.

Q20: Desired changes? A dashboard that shows exactly what they have on me. Not a 50-page privacy policy — like an actual interface showing stored data. Ephemeral mode that's verifiably ephemeral. And third-party audits. Trust but verify, right?

F. Participant P2 – Psychology, Female, 18–24

Demographics:

Q1: How old are you? I'm 21.

Q2: What is your gender? Female.

Q3: What is your highest level of education? Um, I'm in my third year of undergrad, so... working on my bachelor's?

Q4: What is your field of study or work? Psychology! I love it, except for the stats part. [laughs]

General Usage:

Q5: Just ChatGPT. Like, four or five times a week? More during exams honestly, it becomes like my study buddy. I tried Gemini once but I didn't really see the point, and ChatGPT is what everyone uses so...

Q6: School stuff mostly! Like, I'll take a research paper that's super dense and be like 'explain this to me like I'm five.' [laughs] Stats help too — I'm in psych so there's SO much statistics and I am not a math person. Um, and personal stuff sometimes. Like if I'm overthinking something, I'll just... talk to it? Is that weird? It's like having a really patient friend who never gets tired of you.

Q7: Oh! Okay so I was writing this paper on attachment theory, right, and my lit review was a mess. I literally pasted my whole draft intro and was like 'what am I missing, what's wrong with this.' And it pointed out that I was mixing up two frameworks that I thought were the same thing but aren't. I would've lost so many marks if I hadn't caught that. Like, it literally saved my grade. I shared the whole thing though — my thesis, my argument, my name was probably in the document header, I didn't even think about it.

Q8: I worry I'm getting too dependent on it? Like, will I still be able to write a paper without it? That's my main thing. Um... privacy, not really? I mean, I'm not putting my SIN number in there or anything. [pause] Should I be worried?

Mental Models:

Q9: Like... the cloud? I don't really know what that means technically. Somewhere on the internet? I can see my old conversations on the sidebar so it's saved somewhere. Beyond that, I have no idea. Sorry, I'm not very techy.

Q10: Just me? I mean... I hope just me. [laughs nervously] Can other people see it? I guess the company could if they wanted to? I never really thought about it. I kind of assumed it was like a private diary. Oh god, is it not?

Q11: Hmm, I think I've heard people say that? Like on TikTok or something. But I don't actually know if it's true. It would make sense though — how else do they make it better? I just... I never really looked into it. If they are using my stuff, I guess that's fine as long as it's not like, connected to my name.

Q12: I honestly have zero idea. Forever? Until I delete it? I've never deleted a conversation actually, I didn't even know you could until my friend showed me like a month ago. There's just... a lot of stuff in there at this point.

Disclosure Behaviors:

Q13: Okay yeah, so there was this time when things at home were really rough — my parents were going through some stuff and it was affecting everything, my grades, my sleep, everything. And I just... vented to ChatGPT. Like, the whole story. Family dynamics, my living situation, how I wasn't sleeping. It felt safe because it's not a person who's gonna judge me or like, gossip about it. [pause] But yeah, looking back, that's a lot of personal information just... sitting there.

Q14: The family stuff, kind of. Not exactly regret but more like... afterwards I was like 'huh, that's literally in a database somewhere now.' And once I asked about a medication and described my symptoms in detail. Then I thought — what if somehow that health info gets connected to me? Probably paranoid but still.

Q15: Oh my god, yes. I'll literally be like 'I'm a third-year psych student at uOttawa' because it gives context and I don't even think about it. Or I'll mention my neighborhood when asking for food recs. I never thought of those as personal information until you just asked me. That's actually kind of scary? Like how much am I giving away without realizing...

Q16: Yeah, once I wanted to ask about a mental health thing and I made it hypothetical — like 'if someone was dealing with X, what would...' instead of saying it was me. But that was more because I felt embarrassed, not because of privacy. Is that the same thing? I don't know.

Privacy Concerns:

Q17: Okay so now that we've been talking about this I'm definitely more concerned than when we started. [laughs] The idea that someone at OpenAI could read about my family problems? Ew, no. And if it got hacked — like, my conversations getting leaked? I would die. But like... on a normal day I just don't think about it. Out of sight, out of mind.

Q18: Nope. Well — my friend mentioned something about turning off training? But I wouldn't know where to find it. I've never been in the settings. I should probably look at that, huh.

Q19: Honestly... nothing. [laughs] I guess sometimes I use vague language instead of specifics, but that's more about being embarrassed than privacy. I don't use any settings or anything. This conversation is making me feel like I should probably start.

Q20: Like a pop-up that says 'hey you're about to share health stuff, just so you know this might be stored.' Like how websites do the cookie thing. And make the privacy settings actually visible — not buried in some menu. I'm not gonna go hunting for it, you know? Make it obvious.

G. Participant P3 – Electrical Engineering, Male, 35–44

Demographics:

Q1: How old are you? 41.

Q2: What is your gender? Male.

Q3: What is your highest level of education? Ph.D. in electrical engineering.

Q4: What is your field of study or work? Electrical engineering — I'm a faculty member working in signal processing and IoT security. About fifteen years in the field.

General Usage:

Q5: ChatGPT Plus, daily, roughly six to eight queries a day. Gemini perhaps twice a week, mainly when I need cross-referencing or something in Google's ecosystem. I've been working adjacent to NLP for about fifteen years so I have a reasonable understanding of the underlying mechanisms — for better or worse, since that understanding sometimes makes me more paranoid, not less.

Q6: Research support — literature summaries, related work suggestions, helping draft manuscript sections. Python code for data pipelines. I teach a graduate course, so I'll use it to generate exam questions or rephrase explanations. The teaching use case is underappreciated, I think — it's quite good at offering alternative framings of a concept.

Q7: I was analyzing the privacy policies of several IoT manufacturers — a meta-analysis of sorts. I fed an entire privacy policy document into ChatGPT and asked it to extract data-sharing clauses. I also shared my analysis framework and preliminary coded data. It flagged several clauses I'd missed. The irony of feeding privacy policies into a system with opaque data practices was not lost on me. [slight laugh] But the productivity gain was real.

Q8: Intellectual property, primarily. When I input unpublished findings or novel methodological approaches, there's a non-trivial risk of that information influencing model outputs. Whether that risk is practically significant is debatable, but from a research integrity standpoint, it's concerning. Accuracy is less of a worry for me personally because I can evaluate outputs against domain expertise.

Mental Models:

Q9: Distributed cloud infrastructure — Azure for OpenAI, likely AWS or GCP for Anthropic. Conversation data in a relational or document database with user associations, a parallel pipeline for training data processing, encryption at rest and in transit as baseline. But encryption is necessary, not sufficient — the real question is key management policy and access control implementation, neither of which is transparent to end users.

Q10: Engineers with production access, trust and safety reviewers, data annotators who may be reviewing conversations for quality or RLHF labeling. Potentially legal or compliance teams. Enterprise accounts may add admin visibility. The point is, while access controls presumably exist, the technical

capability to access any user's data is there for someone with sufficient privilege. That's inherent to the architecture.

Q11: For ChatGPT consumer tier, that's documented in the ToS — conversations are used for improvement unless opted out. Anthropic's position is more nuanced; the API has stronger guarantees than the consumer product. Google — well, Google's data utilization practices are extensive. I'd be very surprised if Gemini conversations weren't feeding into some training or evaluation pipeline.

Q12: User-facing deletion and actual data purging are very different operations. Deletion from the UI probably removes the index or pointer. But the underlying data in backups, in training corpora, in analytics systems — that persists. In distributed systems with regular snapshots, true deletion is technically complex and expensive. My estimate is months to years for derived data, even post-deletion.

Disclosure Behaviors:

Q13: Unpublished research data is the most sensitive category I've shared. Draft manuscripts, grant ideas, novel experimental results. From an academic priority standpoint, that's quite sensitive. I also once input de-identified student assessment data — grades, course, some demographics — to run statistical analyses. In retrospect, even de-identified educational records are FERPA-adjacent territory and shouldn't be in these systems.

Q14: The student data, certainly. The de-identification was minimal — course, grades, demographic subset. Re-identification wouldn't be terribly difficult in a small class. I also pasted an email exchange with a collaborator to get help summarizing our discussion, which contained that person's ideas and opinions. I didn't have their consent to share that. Both incidents reflect a pattern of prioritizing convenience over appropriate data handling.

Q15: Constantly. My query patterns map my research trajectory. The datasets I describe, the methods I ask about, the papers I request summaries of — collectively, they reveal current projects, methodological gaps, even funding sources. I'm acutely aware of this. I continue because the productivity benefit is substantial, but it's a deliberate trade-off rather than ignorance.

Q16: Yes, I've developed certain practices. If I'm working on a novel technique, I'll describe the general problem class without naming my specific approach. I never share active peer review details — that would be a serious breach of academic ethics. For most routine queries, though, I'm probably less careful than warranted.

Privacy Concerns:

Q17: Three primary concerns. First, IP leakage — unpublished work entering training data. Second, data persistence — the gap between 'deleted' and actually purged. Third, the aggregation problem, which I think is underappreciated: individually benign queries, taken together, construct a detailed map of my research, my students, my institution. These companies potentially know more about my research pipeline than my department chair.

Q18: I'm aware of most available options. I've disabled the training toggle on ChatGPT. I know about data export and deletion features, and the API's different data handling terms. However — and this is common — awareness doesn't translate perfectly to action. Convenience often wins, especially at midnight when I'm chasing a deadline.

Q19: Opted out of training on ChatGPT. I use the API for the most sensitive work since the data handling agreement is more favorable. I sanitize student data where feasible. Separate accounts for personal and research use. But these precautions degrade under time pressure, which is when I'm using these tools most heavily. There's a predictable gap between policy and practice.

Q20: Independent auditing of data handling — not self-reported compliance, actual third-party verification. Verifiable data retention with technical enforcement. A conversation classification system — mark certain exchanges as 'do not retain.' And comprehensive data inventory tools so I can see what's attributed to me and audit it. Policy commitments without verification mechanisms are insufficient.

H. Participant P4 – Business/Marketing, Female, 25–34

Demographics:

Q1: How old are you? 29.

Q2: What is your gender? Female.

Q3: What is your highest level of education? I have a master's in marketing.

Q4: What is your field of study or work? Marketing — I'm a marketing manager at a SaaS company.

General Usage:

Q5: ChatGPT is my go-to, I use it most workdays — probably four, five times a day. I started using Claude a couple months ago because a colleague said it was better for longer-form writing, and yeah, I'd agree with that. So Claude maybe twice a week.

Q6: Writing, writing, and more writing. [laughs] Marketing copy, email campaigns, blog outlines, social posts. Also competitive analysis — like 'compare the messaging strategies of X and Y in the CRM space.' Brainstorming campaign themes.

Q7: So last month we were preparing a product launch, and I basically gave ChatGPT our entire product brief — features, target audience, pricing, competitive positioning, the whole thing — and asked it to draft press release options. The output was genuinely good, we used a modified version. But then my CTO pulled me aside and was like 'hey, we should probably be more careful about putting product details into AI tools.' And I was like... oh. [pause] I honestly hadn't thought about it at all until that moment.

Q8: The originality thing bugs me — if everyone's using AI for marketing, does everything start sounding the same? And since the CTO conversation, I've started thinking about confidentiality. But before that? It just wasn't on my radar.

Mental Models:

Q9: Their servers, wherever those are. I can see my history so it's saved somewhere in the cloud. Honestly, it's kind of

like email — I know it's stored somewhere but I don't really think about the specifics.

Q10: Me... and probably the company? I'd assume it's like any tech service — there are probably rules about who can see what internally. I trust that there aren't people sitting around reading my marketing strategies. [pause] I mean, I hope not.

Q11: I've heard that. I feel like there was a headline about it at some point but I didn't read the article. [laughs] Did they change it? I genuinely don't know the current situation.

Q12: I haven't really thought about it. I delete conversations sometimes when the sidebar gets messy, but that's purely an organizational thing. I assumed deleting meant deleting. Does it not? [looks concerned]

Disclosure Behaviors:

Q13: The product brief was the big one — unreleased features, pre-announcement pricing, competitive intelligence we'd spent months developing. I've also used client names and campaign budgets when asking about resource allocation. And on a personal note, I once asked for advice about a tough conversation I needed to have with a direct report about performance issues. Described the whole situation in detail.

Q14: After the CTO flagged it, yes. The competitive analysis especially — that's months of strategic work just... handed over. And I can't un-share it.

Q15: All the time. 'How should I position our SaaS product against [specific competitor]' — that reveals our competitive landscape. 'I need to send a campaign to 50,000 subscribers' — that's our list size. These feel like nothing individually but add them up and it's basically a business profile.

Q16: Only since the CTO talk. Now I try to use placeholder names — 'Product X' instead of our actual name. It doesn't always work because sometimes context matters, but I try.

Privacy Concerns:

Q17: Business confidentiality is the big one now. What if our product strategy influences responses that a competitor gets? That's probably paranoid but... is it?

Q18: Only because my CTO told me about the training opt-out on ChatGPT. I turned it off. Before that conversation, I had no idea those settings existed.

Q19: Placeholder names, turned off training. I'm pushing for an enterprise account at work. But all of this is recent — first year of using ChatGPT, I shared everything without blinking.

Q20: A business mode where you know — like really know — nothing is stored or used for training. Maybe a visual indicator? Green for private, red for 'this may be used.' Something dead simple that I can see at a glance.

I. Participant P5 – Graphic Design, Non-binary, 18–24

Demographics:

Q1: How old are you? 20.

Q2: What is your gender? Non-binary.

Q3: What is your highest level of education? I'm doing my bachelor's right now, second year.

Q4: What is your field of study or work? Graphic design.

General Usage:

Q5: ChatGPT and Gemini, like once or twice a week each? I'm not a heavy user. I mostly pop in when I need something specific.

Q6: Brainstorming — like color palettes, concept ideas, that kind of thing. Writing artist statements, ugh, those are the worst. School essays. And sometimes just for fun.

Q7: I was working on this portfolio piece about mental health awareness and I asked ChatGPT for visual metaphors for anxiety. And we just went back and forth, and I started sharing like, my own actual experiences with anxiety to explain what feeling I wanted the design to capture. And it got really personal really fast.

Q8: AI taking creative jobs. That's the big one for me. Privacy? Nah, not really. I feel like I'm not interesting enough for anyone to care about my data.

Mental Models:

Q9: Somewhere on the internet? [laughs] I genuinely don't know. Servers? The cloud?

Q10: I would hope just me? But... I guess the company can too?

Q11: I've heard people say that and it kinda makes sense but I don't actually know.

Q12: No clue. Probably forever? I've never deleted anything.

Disclosure Behaviors:

Q13: The anxiety stuff I mentioned. Also relationship problems — I've like, fully vented about breakups. And once I went off about a conflict with a professor, like really specific details. It feels like talking to a therapist who doesn't charge $200 an hour and doesn't judge you.

Q14: The mental health stuff makes me feel weird sometimes after. Like, those are things I haven't told my close friends, and they're just... in a database?

Q15: Honestly I never thought about it until right now. But yeah — I mention my school, my program, my professors by name sometimes.

Q16: Sometimes with really personal stuff I'll go 'what would you suggest for someone dealing with...' instead of saying it's me. But that's more of an emotional thing than a privacy thing.

Privacy Concerns:

Q17: Okay honestly after this interview, way more than I had before! The emotional stuff — my mental health, my relationships — if that got leaked or someone at the company read it, I would literally die.

Q18: Nope. Zero. Didn't even know they existed. Wait — there's a way to opt out of the training thing? How?

Q19: Nothing. Absolutely nothing. [laughs] This is bad, isn't it?

Q20: A big, friendly pop-up in plain English that says 'here's what happens to your data.' Not legal jargon — actual human words. And a button before you send something personal that's like 'this is sensitive, please don't keep it.'

J. Participant P6 – IT Management, Male, 45–54

Demographics:

Q1: How old are you? 48.

Q2: What is your gender? Male.

Q3: What is your highest level of education? Master's in information technology management.

Q4: What is your field of study or work? IT management. I manage IT infrastructure for a financial services firm — been in the industry over twenty years.

General Usage:

Q5: ChatGPT and Claude daily, Copilot through our Microsoft 365 enterprise license. I manage IT infrastructure for a financial services firm. Collectively I'd estimate twenty or more interactions per day.

Q6: Infrastructure planning, technical documentation, troubleshooting, stakeholder communications, vendor proposal analysis, automation scripts.

Q7: We had intermittent latency issues on our trading platform. Under pressure, I described our entire network topology to Claude — firewalls, load balancers, subnet configs, the lot — along with error logs and timing patterns. It identified an asymmetric routing issue I'd overlooked. Extremely valuable in the moment. In retrospect... I essentially provided a detailed blueprint of a financial institution's network infrastructure to a third-party AI service. Our CISO would not be pleased if he knew. [dry laugh]

Q8: We operate under SOX, PCI-DSS, various privacy regulations. My use of AI chatbots probably constitutes a data handling practice that hasn't been formally approved by our compliance team. I've been operating in a gray area.

Mental Models:

Q9: Cloud infrastructure — Azure for OpenAI, AWS or GCP for Anthropic. Encryption at rest and in transit is baseline. But the relevant question isn't encryption — it's who holds the keys, under what circumstances is data decrypted for processing or review, and what's the access governance model.

Q10: Authorized personnel — engineers, safety teams, potentially legal under subpoena. From a security perspective, any data transmitted to a third party is data you've lost control of. Full stop.

Q11: Consumer accounts, almost certainly. Enterprise and API tiers have stronger contractual guarantees. But I maintain some skepticism about whether opt-out mechanisms are implemented as thoroughly as advertised.

Q12: User-facing deletion is cosmetic. Actual purging from databases, backups, log aggregation, training pipelines, analytics — that's a different matter entirely. I'd estimate data persists in some form for months to years post-deletion.

Disclosure Behaviors:

Q13: The network topology incident. Server configurations, IP ranges, firewall rules, authentication mechanisms. Professionally — I've shared things that would constitute a security incident if our audit team reviewed my chat history.

Q14: The network topology, without question. If that data were exfiltrated through a breach, it would be a significant security event for my organization.

Q15: Every technical question reveals infrastructure detail. Aggregated, my conversations would provide a comprehensive picture of our tech stack, security posture, vendor relationships, current vulnerabilities. That's a social engineering goldmine, frankly.

Q16: I've developed habits — substituting private IP ranges, genericizing vendor names, describing architectures abstractly. But when I'm troubleshooting a production issue at 2 AM, those habits evaporate.

Privacy Concerns:

Q17: Regulatory compliance risk — we may be violating data handling requirements. Aggregation risk — the totality of my conversations is essentially a security assessment of our organization. Supply chain risk — a breach at these AI providers could expose our architecture to adversaries.

Q18: Thoroughly. I've reviewed enterprise terms, enabled opt-out on consumer accounts, evaluated API data processing agreements. But settings and policies are only as reliable as their implementation.

Q19: Enterprise accounts where available, training opt-out on personal accounts, data sanitization when time permits, no credentials or keys shared. But there's a persistent gap between what I know I should do and what I do when a P1 incident hits at midnight.

Q20: SOC 2 Type II certification specific to conversation data. Published, verifiable retention limits. Contractual commitments that survive acquisitions. Real-time data flow transparency. And frankly, a regulatory framework that specifically addresses AI conversation data. The current landscape is the Wild West.

K. Participant P7 – Nursing, Female, 25–34

Demographics:

Q1: How old are you? 26.

Q2: What is your gender? Female.

Q3: What is your highest level of education? Bachelor's — well, I'm finishing up my nursing degree right now.

Q4: What is your field of study or work? Nursing!

General Usage:

Q5: Just ChatGPT, the free version. Maybe four or five times a week, sometimes more during clinical weeks.

Q6: Studying! Like, 'explain the renin-angiotensin system like I'm not a science person.' Quizzing myself on drug side effects. Understanding lab values. And honestly? Stress management. Clinical rotations are intense.

Q7: During my last rotation, I had a patient with this really complex medication regimen and I didn't understand some of the drug interactions. After my shift I described the patient's conditions and medications to ChatGPT — I didn't use their name, but I gave the diagnoses, age range, the full med list. Then my clinical instructor mentioned we probably shouldn't be putting patient info into AI tools, even without names. And I was like... oh no. [nervous laugh]

Q8: Accuracy — in healthcare, wrong information could be dangerous. And after my instructor's comment, the patient privacy thing is really weighing on me.

Mental Models:

Q9: On OpenAI's servers? In the cloud? I don't really understand the technical side beyond that.

Q10: Me, and probably people at OpenAI if they needed to? I'd hope there are rules about who can see user data.

Q11: I think I saw something about that but I can't remember. If they are... that's concerning given the patient stuff I've described. Even without names, the combination of conditions and medications could be pretty specific.

Q12: I have no idea. As long as I have an account?

Disclosure Behaviors:

Q13: The patient scenario I mentioned is the big one. But I've also talked about my own health — sleep issues, exam anxiety. And I described some really tough clinical experiences where patients didn't make it and I was struggling emotionally. Those were like... real, heavy conversations.

Q14: The patient info — definitely. Now that my instructor flagged it, I realize I probably gave enough detail to violate the spirit of confidentiality even without a name.

Q15: Yeah — I'll mention my rotation site, the type of unit, roughly where the hospital is. When I describe patient scenarios, even 'hypothetically,' the details are pretty specific.

Q16: Since my instructor's comment, yeah. I try to make patient stuff more generic now — change the age, combine aspects of different cases. Before that? I just described things exactly as they were.

Privacy Concerns:

Q17: Patient confidentiality is number one — I could face serious professional consequences. Like, that's a PHIPA issue, potentially.

Q18: Not really? I know I can delete individual conversations but I didn't know about other settings.

Q19: Since my instructor's warning — modifying patient details, changing ages, mixing conditions from different cases. Not using names or dates.

Q20: A healthcare mode where data is guaranteed not stored — something that complies with health privacy laws. Automatic warnings when the system detects clinical or health information. And honestly, training for healthcare students on how to use these tools appropriately. Nobody told us about this.

L. Participant P8 – Computer Engineering, Male, 18–24

Demographics:

Q1: How old are you? 22.

Q2: What is your gender? Male.

Q3: What is your highest level of education? Almost done my bachelor's — fourth year, computer engineering.

Q4: What is your field of study or work? Computer engineering. Hoping to go into security after I graduate.

General Usage:

Q5: All of them. ChatGPT daily, Claude for longer coding stuff, Gemini when I need Google integration, Copilot's always on in my IDE. I probably interact with AI more than I talk to real people on some days. [laughs]

Q6: Coding, like 70% of the time. Debugging, writing code, understanding algorithms, assignments. The rest is random — gaming stuff, meal planning, life advice.

Q7: My capstone project — smart home IoT security system. Got stuck on the auth module, so I basically gave Claude... everything. Repo structure, my code, threat model, system architecture. We went back and forth for like an hour on mutual TLS implementation. It was sick. But yeah I basically handed over our entire project. My teammates don't know I did that.

Q8: Academic integrity, mostly. Where's the line between tool and doing-your-homework-for-you? Privacy? Nah. I'm 22, I don't have secrets. What are they gonna do with my debugging sessions?

Mental Models:

Q9: Their cloud. Some DB, probably replicated across regions. Standard stuff.

Q10: Me, and anyone at the company with the right DB access. There's probably RBAC and audit logs, but if a rogue employee or a good enough attacker wants in, they're getting in. That's every online service though.

Q11: Yeah, unless you opt out. ChatGPT uses convos for RLHF and fine-tuning. Claude — Anthropic says they don't, but I'd want to see the infra to fully believe it. Google uses everything, that's just what Google does.

Q12: Indefinitely, basically. 'Delete' probably means 'remove from UI' not 'purge from all backups.'

Disclosure Behaviors:

Q13: Capstone project, internship code under NDA, and... okay this is kinda bad, I once pasted a conversation I had with a friend to ask if I was being unreasonable in an argument. Like, their actual messages. Their words, not just mine. And relationship stuff, with enough detail to be a whole season of a show.

Q14: The friend's conversation — they definitely didn't consent to having their messages sent to OpenAI. That felt wrong after. The NDA code too, technically. But honestly every dev I know does the same thing, so it's like... normalized at this point.

Q15: Yeah, all the time. My code questions reveal my stack, my projects, my internship company. Food recs reveal where I live. But I genuinely don't care about any of that.

Q16: Almost never. One exception — I found a security vuln in an open-source library and I didn't want exploit details in training data, so I kept it vague. That's more about public safety than personal privacy.

Privacy Concerns:

Q17: For me personally? Basically zero. The risks feel abstract, the productivity gains are concrete. The thing I do think about is security-related conversations ending up in training data.

Q18: Yeah, know about them, haven't used them. Don't care enough to toggle anything off.

Q19: Nothing on purpose. Different email accounts for different services, but that's organization not privacy. No sanitization, no privacy modes, no deletion.

Q20: I'm already comfortable, so... [shrugs] But for the industry — better breach protections. If these companies get hacked, the conversation databases are insanely valuable. So better security infra, mandatory breach notification, and maybe liability for data compromises.

M. Participant P9 – Social Work, Female, 35–44

Demographics:

Q1: How old are you? I'm 38.

Q2: What is your gender? Female.

Q3: What is your highest level of education? I have a master's in social work.

Q4: What is your field of study or work? Social work — I've been practicing for about ten years now.

General Usage:

Q5: Just ChatGPT, the free version. I use it maybe once or twice a week? A friend showed me about a year ago. I'm not very techy at all.

Q6: Personal things mostly. Advice on situations, help writing professional emails. I'm a social worker so the job is emotionally heavy, and sometimes I use it for... I guess processing? Like debriefing after a hard day. And occasionally work things, like drafting case notes or finding resources for clients.

Q7: I had this really difficult case with a family in crisis. After the session I was emotionally drained and I just... opened up. Described the family situation, the dynamics, what I was struggling with professionally, how it was following me home. It was cathartic, honestly. It gave me grounding techniques, helped me think through next steps. But I shared a lot about the case. A lot.

Q8: I worry about relying on it instead of proper clinical supervision. And accuracy — it's not a licensed professional even if it sounds like one sometimes. Privacy... I hadn't really thought about it before you contacted me about this interview.

Mental Models:

Q9: On the internet somewhere? I'm sorry, I really don't understand the technical side. I know it's not on my phone because I can use it on my laptop too. So there must be a computer somewhere keeping everything.

Q10: I... I assumed it was just me. [long pause] But the company can see it too, can't they? Oh. That's... I've shared client details. Not names, but... oh. [sounds distressed]

Q11: I had no idea that was even a possibility. Is that something they actually do? Because if so... the things I've described about my clients... [pause] I feel sick about that actually. Those are vulnerable people — children in some cases.

Q12: I genuinely don't know. I never thought about it. I've never deleted anything. Is there a way to delete things?

Disclosure Behaviors:

Q13: The case debriefing — family structures, types of abuse or neglect, children's ages and behaviors, what interventions we were considering. No names, but very specific details. And about my own life — marriage challenges, financial stress, burnout. There were stretches where ChatGPT felt like the only space where I could be completely honest without consequences.

Q14: The client details. I'm realizing right now in this conversation how serious that is. I work with vulnerable families, children. Even without names, the specifics could identify them in a small community. I have professional and ethical obligations that I... [pause] I think I've been careless. That's hard to admit.

Q15: I mention my workplace type, the community I serve, the kind of agency. When I describe cases, the combination of issues and demographics is probably identifiable to anyone who knows the community. I wasn't thinking of any of that as sharing information.

Q16: Rarely. I treat it like talking to a colleague behind closed doors. I might soften things about my own life, but for client cases I give the full picture because I want the best guidance. [pause] This conversation is showing me I need to completely change my approach.

Privacy Concerns:

Q17: Client confidentiality — that's now my biggest concern by far. I have professional obligations under privacy legislation and my code of ethics. If it came out that I'd been describing client situations to an AI tool... [pause] I don't even want to think about the consequences.

Q18: No. I had no idea. I just type and send. You're telling me there's a way to delete everything? I need to do that. Especially the conversations about clients.

Q19: None. I didn't know there were steps to take. I never used real client names but as we've talked about, the details are enough. I feel like I need to educate myself about this immediately. This has been a real wake-up call.

Q20: Mandatory warnings when you might be sharing information about other people — especially vulnerable people. A clear, simple guide at signup about what happens to your data, in normal language. And a professional mode for people who deal with confidential information — counselors, social workers, doctors — where nothing is stored. Period. I also think our professional training programs need to address this. Nobody warned us.

N. Participant P10 – Data Science, Male, 25–34

Demographics:

Q1: How old are you? 31.

Q2: What is your gender? Male.

Q3: What is your highest level of education? Master's in data science.

Q4: What is your field of study or work? Data science — I work at a startup doing data science and ML engineering.

General Usage:

Q5: ChatGPT Plus, ten-plus times a day — it's open all day basically. Claude Pro about five times a day, really like it for data analysis back-and-forth. Copilot in the IDE. Data science at a startup, these are just part of how I work now.

Q6: Data analysis code — pandas, SQL, viz scripts. Model architecture brainstorming. Documentation that nobody wants to write. Pipeline debugging. Draft comms for Slack and email.

And personally — financial planning, health stuff, random rabbit holes at 1 AM.

Q7: I was building a churn prediction model and I shared a sample of our actual customer dataset with Claude. About 500 rows — demographics, usage patterns, subscription info, churn labels. Real customer records. I wanted feature engineering help and the conversation was incredibly productive. [pause] But those were real people's data. Real customers who didn't consent to their records going to Anthropic. I told myself '500 rows is a small sample' to feel better about it. That reasoning doesn't hold up.

Q8: Hallucinations in data analysis — if you're not careful, a confident but wrong statistical interpretation can send you down completely wrong paths. And increasingly, the ethics of feeding customer data into these systems. Our privacy policy to customers says nothing about AI chatbots. We're probably in violation.

Mental Models:

Q9: Azure for OpenAI, likely AWS for Anthropic. Conversation logs in a structured DB, embeddings in vector storage, user accounts in relational, separate training pipeline. Multiple copies across regions for redundancy. I build similar data infrastructure professionally, so this is intuitive.

Q10: The user, authorized company staff, government with valid legal process, and whoever breaches the system. RBAC and audit logs presumably exist, but effectiveness depends entirely on implementation. These companies are high-value targets.

Q11: Consumer tiers, yes, with opt-out available. Pipeline probably includes annotation, RLHF, maybe direct fine-tuning. API and enterprise tiers have better isolation guarantees. I opted out on ChatGPT but I'm on consumer Claude and haven't actually checked those settings. So that's a gap in my own practice.

Q12: Deletion removes the user-facing reference. Derived artifacts — training sets, embeddings, analytics aggregates — persist much longer. True deletion in distributed systems with regular snapshots is practically infeasible without serious engineering investment. Months to years for derived data is my estimate.

Disclosure Behaviors:

Q13: The customer dataset. Also proprietary ML architectures, internal business metrics — revenue numbers when asking for benchmarking. Personal side: my entire investment portfolio when I wanted financial analysis, and a pretty detailed health concern once. In each case, the convenience calculus won over the privacy calculus.

Q14: The customer data, absolutely. Those are real individuals' records — demographics, behaviors, financial indicators — sent to a third party without consent or legal basis. My manager doesn't know. The investment portfolio made me uncomfortable too. But the customer thing keeps me up at night sometimes.

Q15: Constantly. ML architecture questions reveal our technical approach. Data challenge questions reveal our maturity gaps. Tool-specific questions map our stack. Financial questions reveal income, risk tolerance, goals. The signal leakage is significant and continuous.

Q16: Since the customer data incident, I've improved. I now generate synthetic data that mirrors the statistical properties of real data. For proprietary code, I try to abstract the problem first. Personal questions — still share freely.

Privacy Concerns:

Q17: Three levels. Professional liability — sharing customer data without authorization, potential PIPEDA violation. Competitive intelligence — technical approaches and business metrics sitting in someone else's infrastructure. Personal aggregation — across hundreds of conversations, these companies have a remarkably complete profile of my professional and personal life.

Q18: Yeah, very familiar. Opted out on ChatGPT, periodic data exports to audit what's stored, aware of deletion features. Claude — I know about conversation deletion but haven't dug into the data handling settings. Which is hypocritical given that I just sent customer data there.

Q19: Training opt-out on ChatGPT. Synthetic data for work tasks now. Abstracting problems before sharing proprietary details. Keeping sensitive financial chats isolated for later deletion. Pushing for API access with enterprise DPAs at work. But this is all recent — first year, I shared absolutely everything without restriction.

Q20: Verifiable data handling through technical mechanisms — zero-knowledge architectures, client-side encryption — not just policy documents. Clear liability frameworks for data compromises. Industry standards for AI conversation data, analogous to PCI-DSS. And automated DLP at the input layer — detect and flag sensitive data before it's sent.

Table II presents descriptive statistics for all Likert-scale survey items.

TABLE II
DESCRIPTIVE STATISTICS FOR SURVEY LIKERT-SCALE ITEMS (N=47)

Item	M	SD
General Perceptions (1=Strongly Disagree, 5=Strongly Agree)		
Comfortable sharing personal details	3.13	0.82
Handle information appropriately	3.30	0.83
Concerns about data handling	3.40	0.77
In control of shared information	3.38	0.77
Privacy Knowledge (1=Strongly Disagree, 5=Strongly Agree)		
Providers store history by default	3.51	0.80
Conversations used for training	3.64	0.85
Can delete conversation history	3.62	1.03
Third parties may access data	2.60	1.10
Providers share data with companies	2.34	1.05
Disclosure Intentions (1=Very Unlikely, 5=Very Likely)		
Cover letter (work history)	3.94	0.87
Health symptoms	3.85	1.10
Restaurant (neighborhood/address)	3.47	1.06
Relationship conflict	3.43	0.97
Proprietary work document	3.70	0.93
Financial advice (income/expenses)	3.11	1.07

Table III summarizes the inferential statistical tests conducted.

TABLE III
INFERENTIAL STATISTICS SUMMARY

Test	Statistic	p-value	Sig.
Spearman's Correlations			
Priv. Knowledge vs. Disclosure	ρ=0.303	0.038	*
Priv. Concerns vs. Disclosure	ρ=0.102	0.494	n.s.
Mann-Whitney U (Technical vs. Non-Technical)			
Privacy Knowledge	U=227.5	0.363	n.s.
Disclosure Intentions	U=213.5	0.224	n.s.
General Perceptions	U=291.0	0.654	n.s.

*$p < 0.05$; n.s. = not significant

Both researchers independently coded transcripts from P1 and P3 (approximately 20% of the data). Inter-coder reliability was assessed using Cohen's Kappa ($\kappa = 0.78$), indicating substantial agreement. Eight coding discrepancies were identified, discussed, and resolved collaboratively. The remaining eight transcripts were divided (Fangyuan: P5, P7, P9; Jingyan: P2, P4, P6, P8, P10).

Initial open coding produced 47 codes, which were grouped into 12 candidate themes and refined to 4 final themes:

Theme 1 codes: productivity justification, time pressure override, conscious risk acceptance, convenience calculus, emotional urgency, work deadline pressure, cost-benefit rationalization.

Theme 2 codes: vague cloud model, private diary assumption, technical accuracy without behavioral change, training data awareness, deletion uncertainty, access control naivety, data persistence underestimation.

Theme 3 codes: location leakage, institutional identification, third-party disclosure without consent, aggregation awareness, professional context normalization, indirect information sharing, query pattern revelation, emotional disclosure escalation.

Theme 4 codes: intention–action gap, trigger event required, setting unawareness, passive avoidance preference, externally motivated behavior change, knowledge without action, inconsistent protective practices.